Supplementary material for

Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations

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S1 Transfer memo: what is prior work and what is new

Written before any code was produced. Every row in the first column is prior work and is cited rather than claimed.

Transfers unchanged (prior work)	Transfers with modification (new here)	Does not transfer
Evidence record (P, S, A, L): provenance set, typed support channels, applicability scope, complete-source lineage	The domain of D_y becomes a *temporal* required-source set over half-open sample intervals	Polyline simplification in every form; there is no geometric tolerance in audio
Provenance as a set of ancestry atoms, mixed produced only by union	Alphabet changes from measured/interpolated to captured/generated; the measurand is different even though the algebra is identical	Terrain support profiles, directional ray classifiers, roughness proxies
Same-channel meet by minimum	Unchanged as an operation; the channel is capture-support	Contour extraction rules and closed-ring anchoring
Absence of verified evidence is absorbing and is not an atom	Strengthened: a $\perp$ element also makes every numeric channel unavailable over the represented span (requirement iv′)	Every spatial result, table and figure
Applicability intersection and lineage union	Unchanged	The spatial simplification operator itself
Non-promotion in the spatial setting	Restated modality-independently and, following the Phase-0 search, presented as an *instance* of a known meet-semilattice construction rather than as a new theorem	Any prior source-code audit and its counterexample

Transfers unchanged (prior work)	Transfers with modification (new here)	Does not transfer
Conformance-test philosophy: exhaustive enumeration, deterministic adversarial fixtures, a separately implemented oracle, frozen expectations, explicit non-prevalence language	Reused as a method template; all results are new	Any prior third-party generalizer audit
—	**New with no spatial analogue:** the kernel-footprint rule with declared per-operator implementation margins and measured model-versus-implementation deviation; footprint monotonicity; signal transparency on decoded PCM; a signed C2PA transport with temporal regions of interest; the unverified state as an operational outcome of a real validator	—

S2 Novelty search and closest-work matrix

Search date First pass 2026-08-19; re-run 2026-09-05 before submission. The re-run is reported below with the first pass, since what a novelty search establishes depends on when it was last attempted.

Databases and interfaces Open-literature web search; arXiv; IACR Cryptology ePrint Archive; ACM Digital Library and LIPIcs listings; the C2PA specification site; OpenSLR.

Query strings (1) `provenance semiring non-amplification derived media operator monotone provenance lattice`; (2) `C2PA content credentials audio provenance derived asset ingredient inheritance research paper 2026`; (3) `"provenance laundering" OR "metadata washing" media provenance attack C2PA`; (4) `audio editing provenance preservation composition proof interval lineage trim concatenate resample formal`; (5) `Denning lattice model secure information flow non-interference applied to media provenance labels 2025 2026`; (6) `audio deepfake detection survey 2026 speech antispoofing ASVspoof 5 generalization`; (7) `audio watermarking soft binding C2PA durable content credentials AudioSeal 2026`.

Outcome The columns *complete-source operator rule* and *C2PA-integrated interval-level enforcement* were empty across every prior row. Because provenance-semiring and information-flow work already yields the non-amplification property abstractly, the theorem is not claimed as new; it is presented as an instance and the contribution is relocated to the executable temporal contract, the transport and the measurements. The full matrix is in `paper/PHASE0_novelty.md` in the repository.

Outcome of the re-run The re-run was conducted to falsify the novelty claim rather than to support it, and targeted the three statements this work depends on: that a derived segment must not carry provenance stronger than its weakest required source; that per-operator kernel support is declared and calibrated rather than assumed; and that the resulting interval evidence is carried in a signed C2PA manifest over temporal regions. No prior work stating any of the three was found. The searches did return the established algebraic material the paper takes as given, which is the outcome the claim predicts: the meet-semilattice construction and the restriction of semiring provenance to monotone queries are prior art and are cited as such. Two 2026 preprints are the closest adjacent work, one on anonymous provenance for verifiably edited audio and one a security analysis of C2PA and its implementations; both were already cited before the re-run, and neither defines an inheritance rule over media time. C2PA itself records multiple sources as ingredients but specifies no rule constraining what a derived asset may claim relative to them, which is the gap this work addresses. Nothing found in the re-run changed a claim in this paper or added a reference; a search that surfaces only works already cited is the result we report, not evidence that none was sought.

S3 Threat-model matrix

Attacker capability	Contract behaviour	Not prevented
Passes a mixed-origin asset through a benign transformation to obtain a captured-only claim	Output claim is the meet over the complete required source set, so it stays mixed	Nothing, if the attacker also controls the signer
Trims away the boundaries so that only captured material is at the edges	Interior sources remain in D_y; claim stays mixed	A trim that genuinely removes all generated material, which legitimately yields a captured-only claim
Relies on a filtered operator so that a generated sample bleeds into a neighbouring output region without appearing in the nominal range	The declared kernel footprint enlarges D_y; over-approximation is safe by Proposition 4 (footprint monotonicity)	An under-declared footprint, which is a conformance failure of that implementation
Strips the manifest and redistributes	State becomes UNVERIFIED	Detection of the stripping as intentional; re-signing under a fresh, dishonest chain
Modifies the asset after signing	Hard binding fails; state becomes INVALID	Attacks on the cryptographic layer itself
Signs a false assertion with a valid key	Nothing. The contract governs propagation, not truthfulness	Everything: a valid signature proves signing, not truth
Records a loudspeaker playing synthetic speech	A captured claim propagates, correct at the sensor layer	The semantic origin; see the replay limitation in the manuscript

S4 EM assertion schema

The assertion label is `io.github.1d-coda.emaudio.evidence`. The payload is:


```
{
  "schema": "https://github.com/1D-Coda/em-audio/blob/main/docs/em-audio-schema-1.0.md",
  "policy": "complete-source",
  "operator": "trim",
  "operatorParameters": { "start": 4800, "end": 43200 },
  "asset": { "sampleRate": 16000, "sampleCount": 38400 },
  "intervals": [
    {
      "regionOfInterest": {
        "type": "temporal",
        "time": { "type": "npt", "start": "0.000000000", "end": "0.612500000" }
      },
      "samples": { "start": 0, "end": 9800 },
      "provenance": ["C"],
      "state": "CAPTURED",
      "support": { "capture-support": 0.9 },
      "applicability": { "capture-support": ["digital-asset"] },
      "lineage": ["urn:emaudio:librispeech:1272-128104-0000"]
    },
    {
      "regionOfInterest": {
        "type": "temporal",
        "time": { "type": "npt", "start": "0.612500000", "end": "1.437500000" }
      },
      "samples": { "start": 9800, "end": 23000 },
      "provenance": ["G"],
      "state": "GENERATED",
      "support": { "capture-support": 0.1 },
      "applicability": { "capture-support": ["digital-asset"] },
      "lineage": ["urn:emaudio:espeak-ng:1.52.0:0042"]
    }
  ]
}
```


Normative fields are `provenance` (`null` means unverified and is never an atom), `support`, `applicability` and `lineage`. The `state` field is a presentation label. A channel omitted from `support` is unavailable, which is different from a channel present with a low value. Temporal ranges follow the C2PA `npt` convention with inclusive start and exclusive end; sample-exact bounds are carried alongside because `npt` strings are lossy at sample resolution.

S5 Composition manifest wiring

Two implementation notes for the C2PA-native composition fixtures (manuscript Section on heterogeneous composition), recorded so a replicating reader does not lose time on them. First, the official tool resolves `ResourceRef` identifiers relative to the directory of the manifest-definition file, so an ingredient's stored `manifest_data.c2pa` must be referenced by a path relative to that directory; an absolute path is reported as `resource not found` even when the file exists. Second, a `c2pa.placed` action must carry a hashed-URI reference to its `componentOf` ingredient assertion (C2PA 2.4 §18.16.3);

writing the reference by hand is impossible before signing, and the supported route is to give the action an `ingredientIds` parameter naming the ingredient's `instance_id`, which the claim generator resolves to the hashed URI at signing time. A frozen composition manifest with both `componentOf` ingredients, both resolved `c2pa.placed` references, both source `digitalSourceType` values and the EM aggregate is committed at `fixtures/expected/manifest_composition_componentOf.json`.

S6 Robustness-arm materials

The neural arm uses Piper `piper-tts` 1.7.0 (MIT, VITS architecture) with the voice `en_US-ljspeech-medium`, whose training data is the LJ Speech Dataset. That dataset states "This dataset is in the public domain in the US (and most likely other countries as well)" and consists of LibriVox recordings by Linda Johnson of non-fiction texts published between 1884 and 1964, all public domain. The `lessac` voices were considered and rejected: their Blizzard 2013 training data carries a restrictive licence. No voice of any identified living person is cloned.

The model is fetched rather than committed, by `tools/fetch_voice.sh`, which verifies both files against pinned digests. For `en_US-ljspeech-medium.onnx` the SHA-256 is `6f52a751e2349abe7a76735eb09dc1875298c77ea2342ffd2fef79ff81b87f22`, and for `en_US-ljspeech-medium.onnx.json` it is `141d612cc0a95ed7efc1ca936b845c2364967f2e9217c5dbfcf69fc4d6c65860`. Voice output is 22 050 Hz and is resampled to 16 kHz by stock ffmpeg before use.

The noise arm generates pink noise locally, seeded by clip index so the arm is deterministic, and mixes it under the clip at −18 dB. The exact command, as executed:

```
ffmpeg -i <clip> \
  -f lavfi -i "anoisesrc=r=16000:colour=pink:amplitude=0.12:seed=<index>" \
  -filter_complex "[1:a]volume=-18dB[nz];\
                   [0:a][nz]amix=inputs=2:duration=first:normalize=0[out]" \
  -map "[out]" -c:a pcm_s16le <output>
```

The noise is modelled as a second generated-derived source covering the whole clip, so under the contract every captured segment becomes mixed ancestry while the generated segment stays generated, union adding no new atom. Both facts are asserted per interval by the experiment.

S7 Reproduction

```
git clone https://github.com/1D-Coda/em-audio \
  && cd em-audio \
  && git checkout v1.0.2 \
  && ./run_all.sh
```

`run_all.sh` fetches the checksum-pinned corpus, generates the test signing credential, runs the named regression tests and experiments A–I, regenerates every table, macro and figure from the machine-readable results, writes `results/PREFLIGHT.txt`, and exits non-zero if any conformance check fails. The preflight report is the single source of every number in the manuscript; a checker script verifies that no numeric result appears in the manuscript source outside the generated macro file.

What to record The environment: the person or group, the machine, operating system, hardware, runtime and tool versions, and the release tag and commit reproduced. The execution: a clean clone, dependency installation, a full `run_all.sh`, and a comparison of the emitted machine-readable results against those shipped with that tag.

Which outputs must match, and which must not The classification matters more than the checklist, and it is applied by `tools/verify_reproduction.py` rather than by judgement after the fact.

The *deterministic* outputs are those a correct reimplementation must match exactly: conformance counts and pass/fail totals, promotion classifications, lineage sets, interval structure, support-channel behaviour, and decoded essence digests. Any difference among them exits non-zero.

The *environment-dependent* outputs are those that should be expected to differ, and whose difference is not a defect: wall-clock timings and their interquartile ranges, the absolute cost figures of Section 7.10 of the manuscript, and the bytes of a freshly signed file, which differ because signing incorporates a timestamp. A reproduction reporting identical timings would be the surprising result, not a reassuring one.

A difference in the first class should be classified rather than absorbed, as an environment difference, a tool-version difference, an ambiguity in the specification of the contract, or a genuine discrepancy, and reported as such whichever it turns out to be.

S8 Independent reproduction: environment and per-operator detail

The run reported in Section 7.11 of the manuscript was performed by D. A. Balderrama-Alvarez (Universidad de Sonora, ORCID 0009-0002-5180-0406), who is not an author. His environment, as his own preflight report records it: a 13th Gen Intel Core i7-1355U with 12 logical processors and 8 GiB, Linux 6.6.87.2 on x86_64, Python 3.14.4, FFmpeg 8.0.1-3ubuntu2, `c2patool` 0.27.15, Node v22.22.1 and eSpeak NG 1.52.0. The reference machine for every other number in the paper is macOS 26.5.2 on arm64 with Python 3.11.15, FFmpeg 9.0.1, `c2patool` 0.27.2 and Node v26.7.0. He ran the package twice on consecutive days. The first run was incomplete: two Python dependencies were missing from the instructions he was given, so the figure and robustness arms did not execute. The second run, on 2026-08-27, is the one released and tabulated here; both runs returned the same values for the two operators whose declarations his build does not satisfy. His unmodified outputs, his preflight report, his full pipeline log and the classified comparison his run produced are released in `results/independent/`.

Table S3 gives the containment result per operator on both machines rather than only the operator that differs, so the selection made in the main text can be checked. 3 of the 4 operators that declare a non-zero footprint reproduce their measured reach exactly across the two builds; the MP3 encoder does not, and its declared footprint does not contain the reach his build measures. The other 3 rows declare no footprint and measure no reach on either machine.

S9 Raw rows behind two summary figures

A coefficient of variation and a maximum measured reach are both summaries, and a summary that ships without the rows it summarises has to be taken on trust. These are those rows. Neither table is typed: both are emitted by `tools/make_tables.py` from the same result files the main text reads.

Table S3: Kernel-support containment on the reference machine and on the independent reproducer's machine. *Declared* is the footprint of Table 3 of the manuscript, in source samples; *Reach* is the widest measured influence of a single input sample; *Outside* counts output samples whose dependency falls outside the declared support. The dagger marks the one operator whose reach did not reproduce.

Operator	Declared	Reach (ref.)	Reach (ind.)	Outside (ref.)	Outside (ind.)
silence removal	4 096	3 277	3 277	0	0
time stretch 1.10	2 577	1 906	1 906	0	0
transcode mp3†	2 304	1 555	4 317	0	110
resample 16 8	97	32	32	0	0
normalize	0	0	0	0	0
transcode flac	0	0	0	0	0
trim 10 90	0	0	0	0	0

Table S4: The benchmark of Section 7.10 as nine independent processes. The coefficient of variation is the sample standard deviation over the mean, in percent, with Bessel's correction. The absolute columns move with machine state; under the within-machine variation measured here, the ratio is markedly more stable than either of them.

Process	EM (ms/audio-min)	Baseline (ms/audio-min)	Ratio
1	0.9209	0.2752	3.3466
2	0.9053	0.2709	3.3421
3	0.9106	0.2705	3.3664
4	0.9018	0.2687	3.3569
5	0.9020	0.2693	3.3491
6	0.8994	0.2696	3.3361
7	0.9133	0.2725	3.3514
8	0.8911	0.2662	3.3475
9	0.9012	0.2670	3.3752
CV (%)	0.966	1.015	0.362
measured 2026-08-30T07:58:30Z UTC, load average [2.1, 2.73, 3.38]			

Table S5: Calibration probes per operator, with the context and source position that produced the widest measured influence. The full per-probe record, all 280 rows, is in `results/machine_readable/CALIBRATION.json`; this table gives the worst case for each operator, which is the value the declaration has to cover.

Operator	Probes	Max reach	Worst context	At sample	Declared
silence removal	40	3 277	tone	6 569	4 096
time stretch 1.10	40	1 906	tone	6 537	2 577
transcode mp3	40	1 555	dense	8 192	2 304
resample 16 8	40	32	tone	1 638	97
trim 10 90	40	0	tone	64	0
transcode flac	40	0	tone	64	0
normalize	40	0	tone	64	0

S10 Feasibility log

Two pitfalls in the signed-transport layer are recorded so that a replicating reader does not lose time on them. First, the signing key must be supplied to the official tool in PKCS#8 form and as PEM *content*, not as a path; a SEC1 key or a path produces a decoder error that does not name the cause. Second, a hand-written `c2pa.opened` action without an `ingredients` parameter is rejected with `assertion.action.ingredientMismatch`, because the specification requires that action to carry a hashed-URI reference to the `parentOf` ingredient assertion; the correct move is to let the claim generator insert it.